PANLIN LI (Direct Ph.D. Application)

High-Dimensional Statistical Learning | Data Modeling

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Member, China Society for Industrial and Applied Mathematics

EDUCATION

Xinjiang Agricultural University, School of Mathematics & Physics — Sep. 2023 – Jul. 2027

B.S. in Mathematics & Applied Mathematics · GPA 3.86 / 5.0 · Top 2.5% of major · Advisor: Prof. Hua Huang

- Honors: Outstanding Volunteer (×9) & Outstanding Team, CYL Central Committee; Hematopoietic Stem Cell Donation Honor Certificate; Outstanding Student, Innovation & Entrepreneurship Star (2 consecutive years), Science & Innovation Pioneer Scholarship — XJAU.
- Key outcomes: published 11 papers (8 first-author), with 3 more first-author manuscripts in submission (SCI Q1 Top, SCI Q1, Chinese core); led or participated in 16 student innovation projects (3 national, 6 provincial), one recognized by Huawei Research Institute; 4 utility-model patents and 4 software copyrights as first inventor; 5 national & 7 provincial competition awards; volunteer work covered by *China Daily*, *People's Daily*, and other media outlets.

RESEARCH & PROJECTS

National Statistical Science Research Project · Major Program · Under Application — Multimodal High-Dimensional Statistical Learning for Cotton Yield and Climate Risk

2026 – Present

- Lead-authored the proposal and a 7,000-word survey under advisor guidance; designed a heterogeneous-multimodal tensor-sparse fusion model, a spatio-temporal doubly-debiased Lasso + generalized random forest causal-identification framework, and a multi-stream field-level early-warning scheme. Proposal submitted and under review.

Xinjiang Natural Science Foundation · General Program · Participant — Functional Statistical Modeling via Spatial-Spectral Feature Fusion of Hyperspectral Imaging — **2026 – Present**

- Curated multi-cultivar HSI/NIR datasets with spectral correction and multi-basis functional representation; developed three directions — functional contrastive sparse multi-task learning, mechanism-embedded post-harvest modeling, and residue spectral screening — and lead-authored three corresponding drafts (under advisor review).

National Innovation Project · Ongoing · Participant — Apple Origin Traceability via Machine Learning & Transfer Learning

2025 – Present

- Lead-authored the proposal; led cross-year apple HSI and image data collection across Xinjiang / Shandong / Gansu; built a 9-domain benchmark with an origin × year leakage-controlled evaluation protocol and ran PLS-DA / SVM / 1D-CNN / Transformer baselines; first-author submission, rejected by *Computers and Electronics in Agriculture* and retargeted to *Transactions of the CSAE*.

Other projects: Small-Sample Time-Series Forecasting for Energy Economics (National · Completed · PI) · Wenmai Zhiyuan Hub (Provincial · Excellent · PI) · Digital Cotton Field (Provincial · Ongoing · PI) · Multimodal Navigation for the Visually Impaired (Provincial · Ongoing) · Apple SSC Prediction via FDA + Image Fusion (Provincial · Excellent) · Nonlinear Wave Solutions of High-Dimensional PDEs (University · Excellent · PI), among others.

PUBLICATIONS

- Li P.L., Li Y.C., Huang H., et al. *Functional Identifiability Analysis of Hyperspectral Spatial–Spectral Fusion with Computable Diagnostics*. ISPRS Journal of Photogrammetry and Remote Sensing. (SCI Q1 Top; IF: 12.9; Under Review)
- Li P.L., Li Y.C., Li L.J., et al. *The label economics of calibration transfer: a five-benchmark, dual-modality evaluation of simple corrections versus deep and physics-informed representations*. Chemometrics and Intelligent Laboratory Systems. (SCI Q1; IF: 3.8; To Be Submitted)
- Li P.L., Zhang Q.L., Jiao Q.T., et al. *A leakage-free few-shot evaluation protocol for cross-year apple origin traceability*. Transactions of the CSAE. (PKU core / EI Compendex; To Be Submitted)
- Li P.L., Zhang N. *Optimization of NIPT Testing Timing and Fetal Anomaly Determination Models*. International Journal of Public Health and Medical Research, 2026, 6(2): 1–8.
- Li P.L., Bao F.C., Zhang J.Y., et al. *Periodic Solutions to a New Class of KdV Equations*. Advances in Applied Mathematics, 2025, 14(5): 23–28.
- Li P.L., Zhang N. *Prediction of optimal NIPT timing and analysis of influencing factors of fetal Y chromosome concentration based on robust Bayesian ridge regression*. Academic Journal of Computing & Information Science, 2025, 8(10): 99–107.
- Li P.L., Zhang N. *Study on the determination of fetal chromosomal abnormalities based on multi-model regression*. Academic Journal of Mathematical Sciences, 2025, 6(2): 121–129.
- Li P.L., Zhang N. *Study on the effect of n-hexane insolubles on pyrolysis product yields based on statistical regression analysis*. Highlights in Science, Engineering and Technology, 2024, 116: 270–276.
- Li P.L., Zhang N. *Research on Optimizing High-dimensional Data Model Based on Principal Component Analysis*. In: 2025 IEEE 3rd International Conference on Control, Electronics and Computer Technology (ICCECT), Jilin, China, 2025, pp. 998–1004.
- Li P.L., Li J., Kadir A., et al. *Cultural Heritage and Rural Revitalization: Exploration and Reflections on China's Practice*. Smart Agriculture Guide, 2024, 4(21): 185–188.
- Rao W.J., Li P.L. *An improved kernel regularized nonhomogeneous grey model based on conformable fractional accumulation*. Transactions on Computational and Applied Mathematics, 2024, 4: 137–147. (Co-First Author)

- Zhang N., Li P.L., Bao F.C., Liu X.Y. *Solitary Wave Solutions and Interaction Solutions of an Extended (3+1)-Dimensional Shallow Water Wave Equation*. Journal of Changchun Normal University, 2026, 45(4): 6–11.
- Jiao Q.T., Li P.L., Mei T., et al. *ESP32-Based Multi-Sensor Near- & Far-Range Water Quality Monitoring System*. Internet of Things Technologies, 2026, 16(2): 9–12, 16.
- Huang H., Wang M., Liu Y., Li P.L., et al. *Prediction of Apple Soluble Solids Content Using a Functional Partial Linear Regression Model Fusing Image Color Features and Vis-NIR Spectra*. Spectroscopy and Spectral Analysis, 2025, 45(S1): 415–423.
- **Utility-Model Patents (4, as first inventor):** Section-Connection Component for a White Cane · Portable Ultrasonic Detection Device · Pesticide Spraying Drone · Foldable Anti-Collision Spraying Boom for Drones.
- **Software Copyrights (4, as first author):** Intelligent White Cane Vision Recognition System · Smart Cane Data Collection & Monitoring System · User Behavior Analysis and Recommendation System · Big-Data-Driven Market Analysis and Forecasting System.

Competitions & Awards

- China International College Students' Innovation Competition (2024) · Bronze Award — *National* — 2024.10
- 11th National Undergraduate Statistical Modeling Contest · Third Prize — *National* — 2025.10
- "Chia Tai Cup" National Market Survey & Analysis Contest (15th & 16th) · Third Prize × 2 — *National* — 2025 – 2026
- DataBee Cup National Interview Survey Contest · Third Prize — *National* — 2024.10
- 11th National Undergraduate Statistical Modeling Contest, Xinjiang Division · First Prize — *Provincial* — 2025.08
- "Chia Tai Cup" Market Survey & Analysis Contest, Xinjiang Division · First Prize — *Provincial* — 2025.04
- National Mathematical Modeling Contest, Xinjiang Division · Second Prize — *Provincial* — 2024.11
- Plus 4 further provincial awards (China International College Students' Innovation Competition 2025 Bronze; 14th China Innovation & Entrepreneurship Competition Third Prize; 10th Statistical Modeling Contest Second Prize; Information Literacy Competition Excellence Award).

Student Experience

Class Monitor, Mathematics 2302, School of Mathematics & Physics — 2023.09 – 2027.07

Class Monitor, 19th Youth Marxist Training Program, Xinjiang Agricultural University — 2025.05 – 2026.05

Head, Volunteer Association, School of Mathematics & Physics, XJAU — 2024.09 – 2025.11

- Founded the 20-member "Shield of the Future" volunteer team; took part in 11 public-welfare programs organized by the CYL Central Committee and the China Young Volunteers Association across Shandong, Henan, and Xinjiang; led the class to the university's *Outstanding Class Collective* honor.

Skills

- **Language:** TOEFL 110/120; first-authored multiple English-language papers; able to write and submit academic English independently.
- **Professional:** high-dimensional statistical inference (debiased Lasso, double machine learning, generalized random forests), functional data analysis (FPCA, function-on-function regression), tensor decomposition & multimodal fusion; Python (PyTorch / scikit-learn), end-to-end modeling from theory to implementation to validation.
- **Tools:** LaTeX (this resume is typeset in LaTeX) · Git · Lightroom / Final Cut Pro.

Self-Assessment

- Photographer and outdoors lover with a strong exploratory drive, hands-on ability, and execution. Through 16 projects and 11 papers I have walked the full loop from mathematics → statistics → machine learning → real-world deployment — which made clear that my bottleneck is theoretical depth. In the Ph.D. stage I aim to move from application-driven work toward more theoretical methodology, while staying grounded in real applications.